JUAN FERNANDO PINZON

ML/AI Engineer | Machine Learning in Production | LLM Applications & Agentic AI

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SUMMARY

ML/AI engineer working in machine learning and data science since 2019, operating production forecasting at PepsiCo since 2022 (XGBoost/LightGBM ensembles on millions of multi-country rows weekly) and co-leading the governance, monitoring and retraining design that sustains it. Earlier, applied deep learning research and client proofs of concept in NLP, computer vision and time series; first author of a neural NER paper (2020), co-author of a causal time-series paper (AGI 2022). Procurement domain background since 2014. Now prototyping the agentic AI roadmap for PepsiCo Digital Procurement and building ProcureRAG, an independent, evidence-driven RAG and agent system: hand-built retrieval measured on a 93-query benchmark, a bounded LangGraph agent with human approval and guardrails, FastAPI/MCP serving, eval-gated CI/CD to Azure Container Apps.

TECHNICAL SKILLS

Languages & data: Python (Pandas, NumPy, Matplotlib, Seaborn), PySpark, SQL

ML / DL (production and research, since 2019): PyTorch, scikit-learn, XGBoost, LightGBM, LSTM/RNN, transformers (BERT), ensembles, classical ML; NLP and Named Entity Recognition (NER), time-series forecasting, computer vision

Production cloud & MLOps (PepsiCo, since 2022): Azure Databricks, Azure Data Factory, Azure AI Foundry, model governance, monitoring and retraining pipelines, Power BI

LLM & RAG engineering (hands-on, ProcureRAG 2026): Retrieval from primitives (TF-IDF, BM25, dense embeddings, hybrid RRF, cross-encoder reranking); RAG evaluation (P@k/MRR/nDCG, DeepEval, RAGAS, LLM-as-judge, regression suites); LangGraph/LangChain agents (state graphs, checkpointing, human-in-the-loop); guardrails (OWASP LLM Top 10); LangSmith; OpenAI/OpenRouter APIs; Hugging Face

Serving & delivery (demonstrated in ProcureRAG): FastAPI, MCP, Docker, GitHub Actions CI/CD, Azure Container Apps (OIDC), pytest, uv, Git, Linux/Unix, REST APIs

Automation & apps: n8n, Streamlit

EXPERIENCE

Data Product Management Associate Manager | PepsiCo Jan 2026 – Present

Barcelona, Spain

- Define and prototype the agentic AI roadmap for Digital Procurement — RAG-based knowledge management agents and RaaS solutions — using Copilot declarative agents and PepsiCo's internal AI platform; contribute to solution architecture and requirements definition for retrieval and orchestration design.
- Built Python data pipelines automating HR data refreshes to ensure adoption KPIs are accurately reported to senior leadership across all Digital Procurement initiatives.

Data Science Sr. Analyst | PepsiCo May 2022 – Jan 2026

Barcelona, Spain

- Supported PepsiCo's first Integrated Business Planning (IBP) implementation, operating ML-based forecasting (XGBoost, LightGBM, ensembles) on millions of multi-country rows weekly; co-led the Sustain model design covering governance, documentation, monitoring, and retraining processes. Stack: Azure Databricks, PySpark, Azure Data Factory. (IBP — Iberia, 2022–2024)
- Designed a cross-functional analytics solution integrating Spend, R&D, and BoM datasets and delivered a Power BI dashboard that cut analysis time from 4+ weeks to near-real-time. (Beyond Spend — Procurement Excellence, 2024–2025)
- Co-led the European rollout of a global spend-visibility platform across 100+ procurement members, then took on a global stretch assignment owning transformation, change management, and adoption KPIs. (SpendHQ, 2025)

Machine Learning Researcher / DL Consultant | Novel Software Systems Aug 2019 – May 2022

Novosibirsk, Russia

- Deep learning consultant within a global ML/DL team at a bioinformatics software company, delivering applied AI research and client proofs of concept across biomedical/life-sciences, HR tech, and financial sectors.
- Built and evaluated neural NER models (Flair/PyTorch BiLSTM-CRF sequence taggers with stacked FastText and contextual string embeddings) for automated resume parsing, identifying that NER alone was insufficient without significant post-processing — first-author paper in Procesamiento del Lenguaje Natural (SEPLN), 2020, with a public 545-CV annotated dataset.
- Co-developed ML/DL models (LSTM ensembles, XGBoost, Random Forest) for cryptocurrency market prediction, integrating order book, market, and social-sentiment data; contributed to a PyTorch backtesting framework — published in the AGI 2022 proceedings (Springer LNCS, 2023).

Business / Systems Analyst | ITMS Nov 2018 – Aug 2019
Novosibirsk, Russia

- Gathered requirements and supported QA for blockchain and cryptocurrency projects, translating stakeholder needs into system specifications.

Production Planning Analyst | Aritex de Colombia S.A.S. Aug – Dec 2016
Cali, Colombia

- Established the Production Programming function and automated ERP data extraction and reporting for real-time performance monitoring.

Procurement & Inventory Analyst | STF Group Jan 2014 – Jul 2016
Cali, Colombia

- Progressive roles across master data, inventory, and textile procurement; introduced automated reporting and reduced obsolete inventory by 20%.

PROJECTS

ProcureRAG — Procurement-Intelligence Copilot (RAG + Agents) | Independent Project Sep 2026 – Present
Open source: github.com/juanfpinzon/ProcureRAG (Python 3.11). A production-shaped RAG system that answers procurement policy questions with cited sources over a synthetic 34-document knowledge base; no employer data or affiliation.

- Implemented retrieval from primitives (inverted index, TF-IDF, BM25, dense embeddings, Reciprocal Rank Fusion, cross-encoder reranking) and benchmarked nine configurations on a 93-query labeled set: reranked hybrid retrieval reached P@1 0.978 / MRR@10 0.984 vs. 0.871 / 0.924 for TF-IDF (ranking metrics, not answer accuracy); failures root-caused to individual queries.
- Added an evaluation layer (citation and grounding checks, DeepEval/RAGAS LLM-as-judge, failure taxonomy, CI regression suite) and a bounded LangGraph agent: evidence-gap-triggered second retrieval pass, interrupt-based human approval, checkpointing, LangSmith-linked traces, and mutation-tested guardrails mapped to the OWASP LLM Top 10.
- Served the same guarded pipeline through a FastAPI API (SSE progress events, role-based filtering, rate limiting) and a read-only MCP tool; non-root Docker image; 430+ tests; eval-gated GitHub Actions CI (three required checks); deployed by image digest to Azure Container Apps via an OIDC-authenticated workflow, with a tested rollback (learning deployment, since torn down).

OpenClaw / Hermes — Self-Hosted AI Agent Setup | Personal Project 2026 – Present
Self-hosted on a personal Linux/Proxmox homelab server

- Configured and extended the open-source OpenClaw and Hermes Agent frameworks (credited upstream) with personally authored Python automations: Linear, GitHub and Telegram connectors, an LLM-based pull-request review watchdog, news and newsletter pipelines, a Markdown shared-memory vault and context-engineering conventions for coding agents.

EDUCATION

M.Sc. — Big Data Analytics & AI | Novosibirsk State University (NSU), Novosibirsk, Russia 2017 – 2019

B.B.A. — Business Administration | Universidad ICESI, Cali, Colombia 2015

CERTIFICATIONS & COURSES

- Boot.dev — Learn Retrieval Augmented Generation (certificate of completion, Sep 2026)
- Hugging Face — The MCP Course, Unit 1: Fundamentals of MCP (certificate of achievement, Oct 2026)
- Microsoft Azure AI Engineer Associate (in progress)
- Also: AI Automation — Build LLM Apps & AI Agents with n8n and APIs; Scalable Machine Learning on Big Data using Apache Spark; Machine Learning (Coursera); The Foundation Negotiator; Internal Coach (PepsiCo, 2025)

PUBLICATIONS

- Pinzon, J. F., Krainikovsky, S., & Samarev, R. (2020). "Limitations of Neural Networks-based NER for Resume Data Extraction." *Procesamiento del Lenguaje Natural* (SEPLN), 65, 53–58. First author. Public dataset (545 annotated CVs): github.com/juanfpinzon/resume-dataset
- Kolonin, A., Raheman, A., Vishwas, M., Ansari, I., Pinzon, J., & Ho, A. (2023). "Causal Analysis of Generic Time Series Data Applied for Market Prediction." In *Artificial General Intelligence (AGI 2022) proceedings*, Springer LNCS, pp. 30–39. DOI: doi.org/10.1007/978-3-031-19907-3_4. Preprint (2022): arXiv:2204.12928

LANGUAGES

Spanish (Native) • English (Full Professional) • Russian (Elementary)